

Set 25: Percentage composition

Notes

We are very interested today in knowing the exact composition of many of the products which surround us. When you buy foods, there are consumer panels listing the percentage of the different ingredients present. Medications also have the amount of active ingredient listed. There is increasing consumer demand that producers should supply specific information about goods. Food and medical goods are analysed by chemists to determine the exact percentage compositions of the ingredients. Individual compounds can also be analysed to determine their exact composition.

The percentage composition of a chemical compound specifies the percentage by mass of each of the different elements present in the compound.

$$\% \text{ mass} = \frac{\text{mass of part}}{\text{mass of whole}} \times 100\%$$

Examples

1. Calculate the percentage composition of ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$

% N	=	$\frac{28.02}{132.144} \times 100 = 21.2\%$	M(N)	=	14.01 g mol^{-1}
			M(H)	=	1.008 g mol^{-1}
			M(S)	=	32.06 g mol^{-1}
% H	=	$\frac{8.0}{132.144} \times 100 = 6.10\%$	M(O)	=	16.0 g mol^{-1}
% S	=	$\frac{32.06}{132.1} \times 100 = 24.3\%$	M($(\text{NH}_4)_2\text{SO}_4$)	=	$132.144 \text{ g mol}^{-1}$
% O	=	$\frac{64.0}{132.144} \times 100 = 48.4\%$			
		<u>100 %</u>			

Note: The percentages should add up to 100. The difference in the calculated value is due to rounding.

2. Calculate the percentage of water of crystallisation in iron(II) sulfate-7-water, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

%H ₂ O	=	$\frac{\text{mass}(\text{H}_2\text{O})}{\text{M}(\text{FeSO}_4 \cdot 7\text{H}_2\text{O})}$			
%H ₂ O	=	$\frac{7(18.016)}{278.22} \times 100$	M($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)	=	$278.022 \text{ g mol}^{-1}$
%H ₂ O	=	45.4%	M(H ₂ O)	=	$18.016 \text{ g mol}^{-1}$

3. A sample of soft solder consisted of 4.77 g of tin and 9.54 g of lead. Calculate the percentage composition of the solder.

% Sn	=	$\frac{m(\text{Sn})}{m(\text{solder})} \times 100$
	=	$\frac{4.77}{(4.77 + 9.54)} \times 100$
	=	33.3%
% Pb	=	$100 - 33.3 = 66.7\%$

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Set 25: Exercises

- Calculate the percentage by mass of each element in:
 - sodium hydroxide
 - acetic acid (CH_3COOH)
 - copper(II) sulfate-5-water
 - potassium phosphate
- Calculate the percentage by mass of:
 - chlorine in calcium chloride
 - sulfur in chromium(III) sulfide
 - oxygen in potassium permanganate
 - nitrogen in ammonium nitrate
- Calculate the percentage by mass of water in:
 - sodium carbonate-10-water
 - nickel(II) sulfate-6-water
 - barium chloride-2-water
- An alloy is prepared by melting together 25.44 g of bismuth, 15.36 g of lead and 7.20 g of tin.
 - Calculate the percentage composition of the alloy.
 - Calculate the mass of each metal required to make 150.0 g of the alloy.
- An alloy used in aircraft construction consists of aluminium, copper and magnesium. An 11.34 g sample of the alloy was treated with alkali to dissolve the aluminium, leaving a residue of mass 2.73 g. This residue was treated with dilute hydrochloric acid solution to dissolve the magnesium. The remaining residue had a mass of 0.900 g. Calculate the percentage composition of the alloy.
- A 3.030 g sample of zinc was heated in oxygen to produce zinc oxide. The mass of zinc oxide produced was 3.771 g. Calculate the percentage composition of the zinc oxide.
- A 16.00 g sample of copper(II) oxide was reduced by reaction with hydrogen gas to yield 12.77 g of pure copper. Calculate the percentage composition of the original copper(II) oxide.
- Copper is found in such minerals as chalcopyrite (CuFeS_2) and malachite ($\text{CuCO}_3 \cdot \text{Cu(OH)}_2$).
 - Calculate the percentage by mass of copper in each.
 - What mass of chalcopyrite must be smelted to produce 100 kg of copper?
- Darling Range bauxite contains high levels of the mineral gibbsite ($\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$) with kaolinite clay ($\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$). Calculate the percentage of aluminium and the percentage of water in each of these substances.
- Ilmenite from Capel contains approximately 53.5% titanium dioxide (TiO_2).
 - Calculate the percentage of titanium in titanium dioxide.
 - What mass of ilmenite must be refined to produce 1.00 tonne of pure titanium metal?